Two-Equation Turbulence Models and Effect of Compressibility

Problem

The objective of this study is to model the backward step over a 2D flat plate using a Computational Fluid Dynamics (CFD) solver ANSYS Fluent employing the two equation model. In this exercise, the two-equation turbulence models, standard $k-\varepsilon$, standard $k-\omega$ and SST $k-\omega$, will be used. This simulation will validate the CFD results against theoretical predictions by comparing:

- Pressure coefficient C_p and wall skin friction coefficient (C_f) at the bottom wall of the plate.
- Velocity profiles and Reynolds stress distributions at streamwise positions $x/H = 1$ and $x/H = 6$.

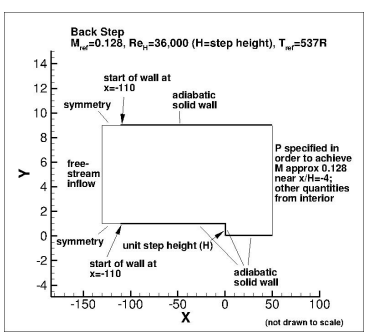
This study not only helps assess the accuracy and limitations of different turbulence models in predicting separated turbulent flows, but also offers insight into model selection for engineering applications involving flow separation and reattachment.

Solution

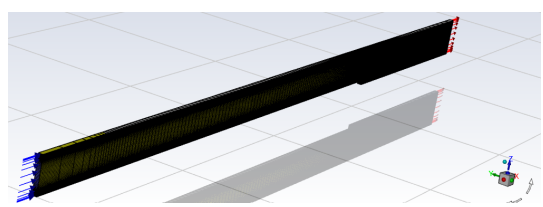
Turbulent separated flows pose a significant challenge in computational fluid dynamics (CFD) due to their complex behavior involving flow detachment, recirculation zones, and reattachment. One of the most widely studied benchmark cases for turbulent separation is the flow over a backward-facing step, which features sudden geometric expansion and flow separation downstream of the step.

This exercise aims to investigate the capability of three common Reynolds-averaged Navier-Stokes (RANS) turbulence models—Standard k–ε, Standard k–ω, and Shear Stress Transport (SST)—to predict flow separation, reattachment, and associated turbulent phenomena in a backward-facing step configuration. The flow conditions, computational grid, and boundary conditions are provided by NASA Langley's turbulence modeling validation database.

The simulations are performed using these three turbulence models, and the results are analyzed in terms of their ability to reproduce key flow features. Boundary conditions along with 2D representation of the plate is presented on the Figure 1a and 3D geometry in ANSYS is presented on Fig. 1b.



(a) 2D surface with boundary conditions, taken from [1].



(b) Mesh representation in ANSYS Fluent.

Setup

At the inlet, a uniform free-stream velocity U_∞ is prescribed to simulate the fully developed channel flow approaching the step. Turbulence quantities (turbulent kinetic energy k and either dissipation rate ε or specific dissipation

rate ω) are also specified based on the selected turbulence model. A pressure outlet boundary condition is applied at the downstream end of the domain, with the gauge pressure set to zero, allowing the fluid to exit the domain without artificial reflection of flow structures. The bottom wall, including the backward-facing step, is modeled using a no-slip boundary condition, enforcing zero velocity at the wall and enabling correct prediction of the boundary layer development, flow separation, and reattachment downstream of the step. The top wall is also treated as a no-slip wall (as in the experimental setup from NASA), to accurately capture the wall-bounded nature of the flow and match the confined channel flow condition. The computational domain is sufficiently extended upstream and downstream of the step to allow for fully developed inlet conditions and reattachment of the separated flow downstream. The mesh is refined in the vicinity of the step and near the walls to properly resolve the shear layer and boundary layer features.

A CFD solver is used to simulate the flow with the following settings:

- Turbulence Model: standard $k-\omega$, standard $k-\varepsilon$, SST $k-\omega$
- Grid: Structured mesh
- Post-processing: Pressure coefficient C_p, skin friction coefficient C_f, velocity and Reynold stress distributions at $x/H = 1$ and $X/H = 6$.

Definition of the quantities are given in the set of equations below.

$$\text{Pressure coefficient: } C_p = \frac{p - p_\infty}{\frac{1}{2}\rho_\infty U_\infty^2} \tag{1}$$

$$\text{Friction coefficient: } C_f = \frac{\tau_w}{\frac{1}{2}\rho_\infty U_\infty^2} \tag{2}$$

$$\text{Reynolds stresses: } -\rho\overline{u_i' u_j'} = \mu_T\left(\frac{\partial \overline{u_i}}{\partial x_j} + \frac{\partial \overline{u_j}}{\partial x_i} - \frac{2}{3}\frac{\partial \overline{u_k}}{\partial x_k}\right) - \frac{2}{3}\rho k \delta_{ij}, \quad \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \tag{3}$$

$$\text{Incompressible flow: } \overline{u'u'} = -\frac{\mu_T}{\rho}\left(2\frac{\partial \overline{u}}{\partial x}\right) + \frac{2}{3}k \tag{4}$$

$$\overline{v'v'} = -\frac{\mu_T}{\rho}\left(2\frac{\partial \overline{v}}{\partial y}\right) + \frac{2}{3}k \tag{5}$$

$$\overline{u'v'} = -\frac{\mu_T}{\rho}\left(\frac{\partial \overline{u}}{\partial y} + \frac{\partial \overline{v}}{\partial x}\right) \tag{6}$$

$$\tag{7}$$

Two-equation Turbulence Models

Standard $k-\varepsilon$

Two-equation turbulence models allow the determination of both, a turbulent length and time scale by solving two separate transport equations. It is a semi-empirical model, and the derivation of the model equations relies on phenomenological considerations and empiricism. The standard $k-\varepsilon$ model is a model based on model transport equations for the turbulence kinetic energy k and its dissipation rate ε. The model transport equation for is derived from the exact equation, while the model transport equation for was obtained using physical reasoning and bears

little resemblance to its mathematically exact counterpart.

In the derivation of the $k-\varepsilon$ model, the assumption is that the flow is fully turbulent, and the effects of molecular viscosity are negligible. This model is therefore valid only for fully turbulent flows. The turbulence kinetic energy, k, and its rate of dissipation, ε, are obtained from the following transport equations [2]:

$$\frac{\partial}{\partial t}(\rho k)+\frac{\partial}{\partial x_i}(\rho k u_i)=\frac{\partial}{\partial x_j}\left[\left(\mu+\frac{\mu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right]+G_k+G_b-\rho\varepsilon-Y_M+S_k$$

$$\frac{\partial}{\partial t}(\rho \varepsilon)+\frac{\partial}{\partial x_i}(\rho \varepsilon u_i)=\frac{\partial}{\partial x_j}\left[\left(\mu+\frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x_j}\right]+C_{1\varepsilon}\frac{\varepsilon}{k}\left(G_k+C_{3\varepsilon}G_b\right)-C_{2\varepsilon}\rho\frac{\varepsilon^2}{k}+S_\varepsilon$$

In these equations, G_k represents the generation of turbulence kinetic energy due to the mean velocity gradients. G_b is the generation of turbulence kinetic energy due to buoyancy. Y_M represents the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate. $C_{1\varepsilon}$, $C_{2\varepsilon}$, and $C_{3\varepsilon}$ are constants. σ_k and σ_ε are the turbulent Prandtl numbers for k and ε, respectively. and are user-defined source terms.

$C_{1\varepsilon}$=1.44, $C_{2\varepsilon}$=1.92, and C_μ=0.09, σ_k=1.0 and σ_ε=1.3 default values have been determined from experiments for fundamental turbulent flows including frequently encountered shear flows like boundary layers, mixing layers and jets as well as for decaying isotropic grid turbulence. They have been found to work fairly well for a wide range of wall-bounded and free shear flows.

The turbulent viscosity, μ_t, is computed by combining as follows:

$$\mu_t = \rho C_\mu \frac{k^2}{\varepsilon} \tag{8}$$

Standard $k-\omega$

The standard $k-\omega$ model in Ansys Fluent is based on a $k-\omega$ model proposed by Wilcox, which incorporates modifications for low-Reynolds number effects, compressibility, and shear flow spreading. One of the weak points of the 1998 Wilcox model is the sensitivity of the solutions to values for k and ω outside the shear layer (freestream sensitivity), which can have a significant effect on the solution, especially for free shear flows [3]. The turbulence kinetic energy, k, and the specific dissipation rate, ε, are obtained from the following transport equations:
In these equations, G_k represents the generation of turbulence kinetic energy due to the mean velocity gradients. G_ω represents the generation of ω. Γ_k and Γ_ω represent effective diffusivity of k and ω, respectively. Y_k and Y_ω represent the dissipation of k and ω due to the turbulence. S_k and S_ω are user-defined source terms. G_b and $G_{\omega b}$ account for buoyancy.

The turbulent viscosity, μ_t, is computed by combining as follows:

$$\mu_t = \alpha^* \frac{\rho k}{\omega}, \qquad \alpha^* = 1.0. \tag{9}$$

$$\frac{\partial}{\partial t}(\rho k)+\frac{\partial}{\partial x_i}(\rho k u_i)=\frac{\partial}{\partial x_j}\left(\Gamma_k \frac{\partial k}{\partial x_j}\right)+G_k-Y_k+S_k+G_b$$

$$\frac{\partial}{\partial t}(\rho \omega)+\frac{\partial}{\partial x_i}(\rho \omega u_i)=\frac{\partial}{\partial x_j}\left(\Gamma_\omega \frac{\partial \omega}{\partial x_j}\right)+G_\omega-Y_\omega+S_\omega+G_{\omega b}$$

Shear-Stress Transport (SST) $k-\omega$

The SST $k-\omega$ model accounts for the transport of the turbulence shear stress in the definition of the turbulent viscosity. These features make the SST model more accurate and reliable for a wider class of flows (for example, adverse pressure gradient flows, airfoils, transonic shock waves) than the above mentioned standard models [4]. Other models fail to properly predict the onset and amount of flow separation from smooth surfaces. The main reason is that other models do not account for the transport of the turbulent shear stress. This results in an overprediction of the eddy-viscosity. The proper transport behavior can be obtained by a limiter to the formulation of the eddy-viscosity:

$$\mu_t = \frac{\rho k}{\omega} \frac{1}{max[\frac{1}{\alpha^*}, \frac{SF_2}{a_1 \omega}]}, \tag{10}$$

where S is the strain rate magnitude and α^* is defined as

$$\alpha^* = \alpha^*_\infty \left(\frac{\alpha^*_0 + Re_t/R_k}{1 + Re_t/R_k} \right), \quad Re_t = \frac{\rho k}{\mu \omega}, \quad R_k = 6, \quad \alpha^*_0 = \frac{0.072}{3}. \tag{11}$$

F_2 is given by

$$F_2 = \tanh \Phi_2^2, \qquad \Phi_2 = max\left[2\frac{\sqrt{k}}{0.09\omega y}, \frac{500\mu}{\rho y^2 \omega}\right], \tag{12}$$

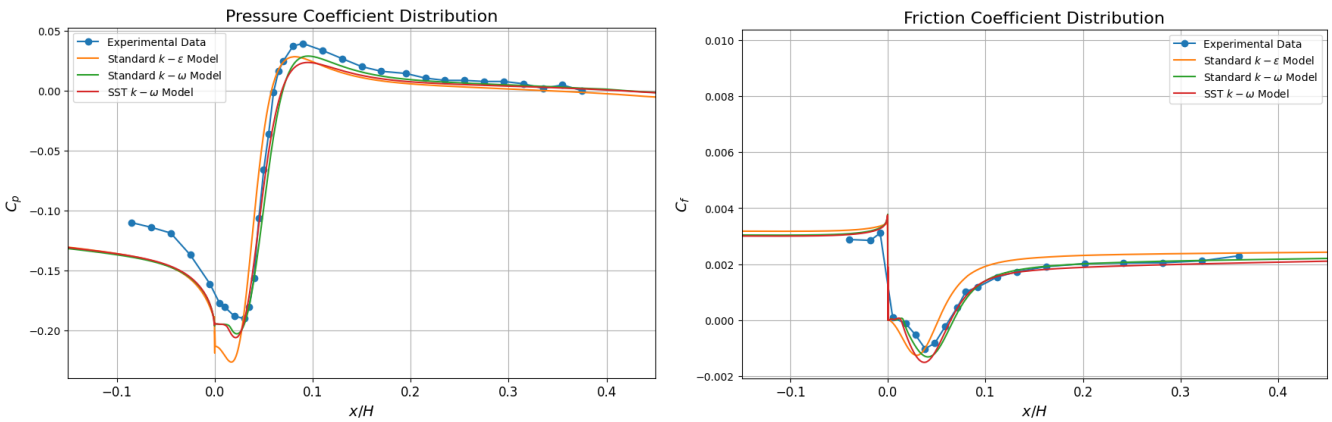
where y is the distance to the next surface.

Results & Discussion

The U_{ref} is the reference velocity at the center-channel near x/H=-4, used to nondimensionalize velocity and turbulent shear stress profiles. The skin friction coefficient and pressure coefficient data were also evaluated with respect to conditions near this location. Note, however, that when plotted the pressure coefficient data has been shifted uniformly so that C_p is 0 near the position x/H=0.4.

Simulation result of the pressure coefficient and friction coefficient are presented on Fig. 2a and 2b, respectively. Just after the step, the flow separates and a low-pressure recirculation zone forms. C_p drops sharply due to local pressure drop. As the flow reattaches, pressure begins to recover downstream. From the Fig. 2a it can be visible that before the step and immediately after the step, theoretical predictions are underestimating the C_p coefficient. In recovering zone all three turbulence models represent physics devotedly comparing to the experimental results. The standard $k-\omega$ model yields the closest match to C_p experimental data in the backward-facing step flow because it offers better near-wall turbulence modeling and accurately predicts the separation-induced pressure drop and recovery. Its robustness in adverse pressure gradients makes it especially suited for flows involving strong recirculation.

From the Fig. 2b can be seen that $k-\omega$ model in this case, as well, shows the best alignment with C_f experimental data. Just after the step, flow separates - shear stress drops to nearly zero or becomes negative (reverse flow at the wall). Sudden increase in C_f indicates reattachment point. In the backward-facing step flow, analyzing C_f allows to assess the ability of turbulence models to resolve near-wall turbulence and predict the correct location of reattachment, which is essential for accurate flow modeling and design. The reattachment point marks the end of the recirculation zone where separated flow returns to the wall. Its accurate prediction is essential for capturing wall shear stress trends and evaluating turbulence model performance in separated flow regimes.



(a) Pressure coefficient validation simulation results with different turbulence models utilized.

(b) Skin friction coefficient validation simulation results with different turbulence models utilized.

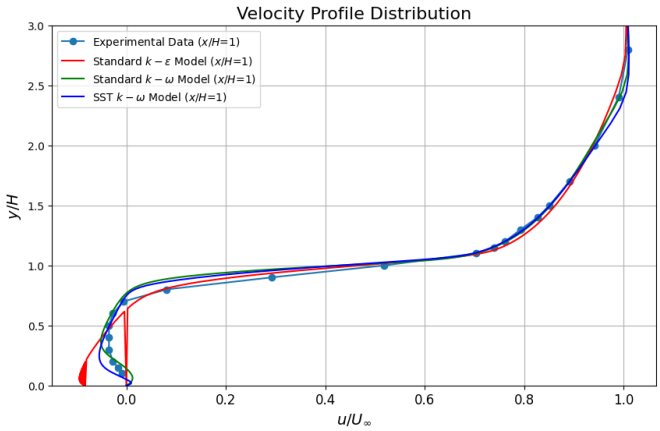
The reattachment points predicted by each simulation are presented in Table 1. Among the models, the SST $k-\omega$ model yields the reattachment point closest to the experimental measurements. The small variations in U_∞, p_∞, ρ_∞ arise due to differences in turbulence model behavior and how each affects the flow field during iterative convergence, even under identical boundary conditions.

Model	U_∞ [m/s]	p_∞ [Pa]	ρ_∞ [kg/m^3]	x/H	C_f
Standard $k-\varepsilon$	44.1383	99851.1	1.16329	0.051667	2.53E-06
Standard $k-\omega$	44.2518	99846.1	1.16325	0.067292	4.06E-06
SST $k-\omega$	44.2431	99846.1	1.16325	0.064583	1.61E-05
Experiment	/	/	/	0.0626 $\pm$ 0.0010	/

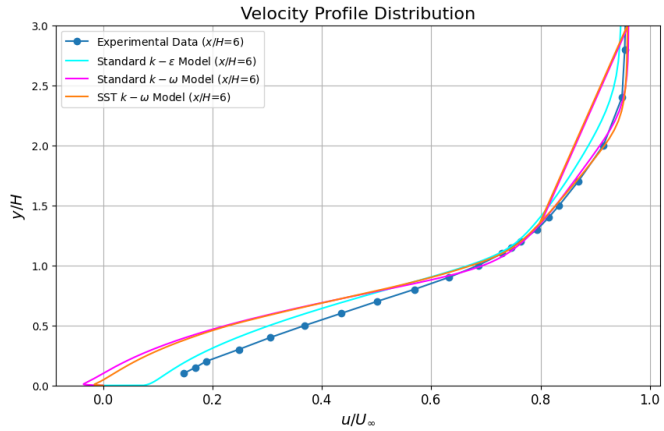
Table 1: Values of U_∞, p_∞, ρ_∞ and reattachment points in simulation cases.

Velocity profiles were observed at x/H=1 and x/H=6. Around x/H=1, depending on the Reynolds number and geometry, the flow begins to reattach. This is a critical zone where: turbulence production peaks, velocity gradients are high, models are pushed to capture both the separation and reattachment dynamics. Comparing velocity profiles and Reynolds stresses here helps assess how well each model captures the separated shear layer and reattachment. By x/H=6 point, the flow has likely reattached and begun to recover toward a new equilibrium boundary layer. This region is useful to check: whether the model can predict the redevelopment of the boundary layer post-reattachment, how well it estimates Reynolds stresses in the fully turbulent flow. It tells you how accurately the model captures the long-term effects of separation. Results are presented in Fig. 3a and 3b.

At the near-reattachment region ($x/H = 1$), the SST $k-\omega$ and standard $k-\omega$ models demonstrate better agreement with the experimental velocity profiles. This is because $k-\omega$ models are more effective at capturing adverse



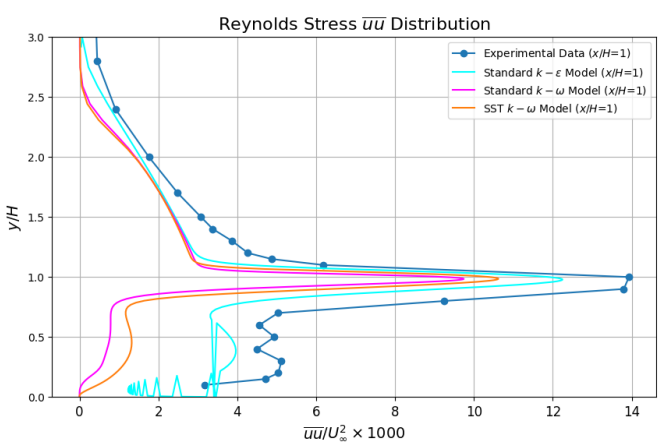
(a) Velocity distribution at $x/H = 1$ for different turbulence models compared with experimental data.



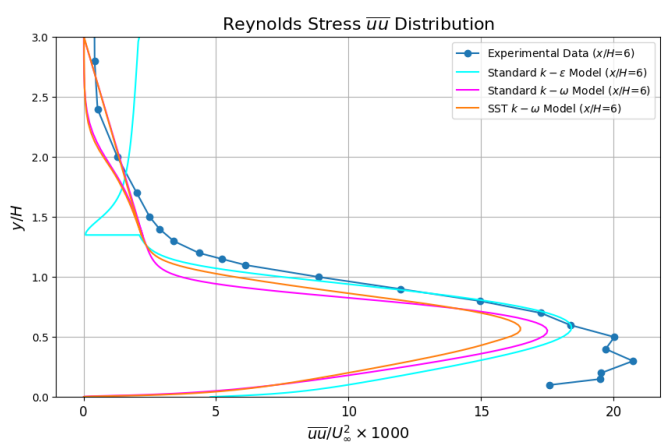
(b) Velocity distribution at $x/H = 6$ for different turbulence models compared with experimental data.

pressure gradients and flow separation, which are dominant in this region. In contrast, further downstream at $x/H = 6$, the standard $k - \varepsilon$ model provides a closer match to experimental data. This can be attributed to the model's strength in predicting fully developed turbulent flows, where the flow has reattached and redeveloped into an equilibrium boundary layer. Unusual fluctuations on the graph are consequences of inadequate generation of file data before postprocessing.

Reynolds stresses represent the turbulent momentum transport in a flow. They are defined as the correlations of fluctuating velocity components, like $\overline{u'v'}$, and are essential in describing how turbulence affects the mean flow. One can simulate Reynolds stresses to evaluate how well turbulence models capture the physics of turbulent transport. In this study, the Mach number is low (Ma$\leq$0.129) — meaning density changes are negligible. The flow is not affected by temperature or pressure changes enough to change density. Most lab-scale backward-facing step experiments and simulations are in low-speed, constant-density conditions. Incompressible assumptions are made because the flow is low-speed, and this simplification doesn't hurt accuracy.



(a) Reynolds stress $\overline{uu}$ component distribution at $x/H = 1$ for different turbulence models compared with experimental data.



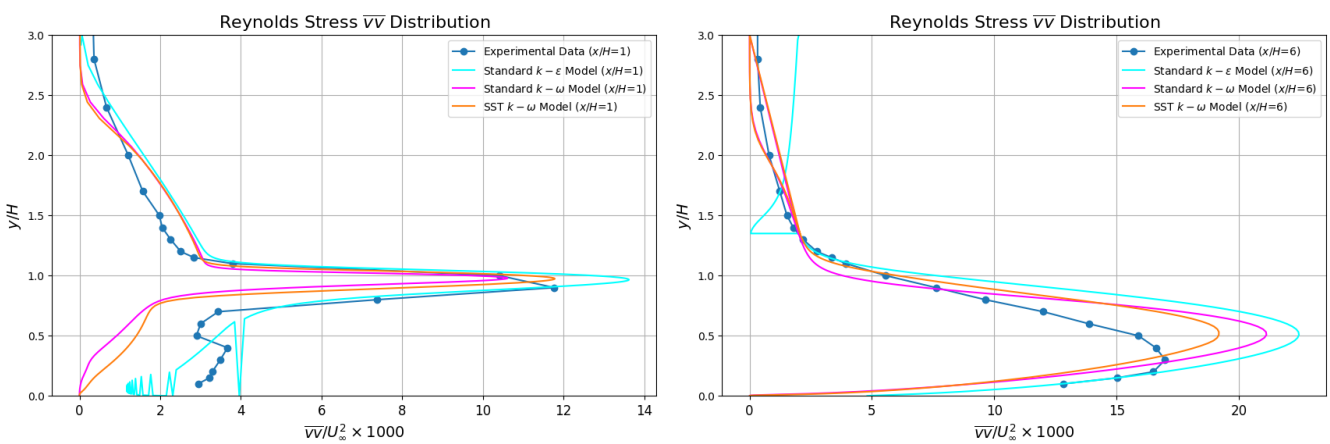
(b) Reynolds stress $\overline{uu}$ component distribution at $x/H = 6$ for different turbulence models compared with experimental data.

Results of Reynolds stress $\overline{uu}$ component distribution are presented on the Fig. 4a and 4b. The results of the Reynolds stress $\overline{uu}$ component distribution are presented in Fig. 4a and 4b. From Fig. 4a, it can be observed that the standard $k-\varepsilon$ model provides the best agreement with experimental data across the entire wall-normal range. However, in Fig. 4b, the same model fails to accurately predict the stress distribution for $y/H > 1$. Sometimes, $k-\varepsilon$ models tend to overpredict turbulence kinetic energy (TKE) in separated flows. This can accidentally align with experimental $\overline{uu}$ profiles in the near-field region, especially when the experimental peak is high. At $x/H = 6$, the flow is reattached and resembles a boundary layer recovering from separation - a tough region for the standard $k-\varepsilon$ model.

Comparing both positions, the $\overline{uu}$ component exhibits a broader distribution at $x/H = 6$, indicating a more diffused turbulent field downstream. At $x/H = 1$, the stress peaks sharply around $y/H = 1$, which is typical near the shear layer formed just after separation. In both cases, the $\overline{uu}$ component remains positive, as it represents the normal stress due to streamwise velocity fluctuations, which by definition cannot be negative.

Results of $\overline{vv}$ component distribution of Reynolds stress is presented on Fig. 5a and 5b. Although the Reynolds stress distributions for $\overline{uu}$ and $\overline{vv}$ components show similar qualitative trends, they represent different physical processes. The streamwise component $\overline{uu}$ is typically larger in magnitude due to stronger velocity fluctuations along the main flow direction. The similarity in shape could be attributed to the isotropic assumptions inherent in the turbulence models, particularly in the standard $k-\varepsilon$ formulation, which may not fully capture the anisotropic nature of turbulence near separation and reattachment zones. At $x/H = 6$, where the flow has reattached and transitioned into a developing turbulent boundary layer, all turbulence models tend to overpredict the peak of the $\overline{uu}$ Reynolds stress. Moreover, the predicted peak is often smeared compared to the experimental data, indicating a broader and less accurate distribution of turbulent energy in the streamwise direction.

This behavior is typical of RANS-based two-equation models, which rely on the assumption of eddy viscosity and isotropy. These models struggle to accurately capture the complex anisotropic turbulence structures and history effects present in reattaching flows. As a result, they tend to diffuse the turbulence too much, leading to overpredicted and smeared Reynolds stress peaks, particularly downstream of the reattachment point.

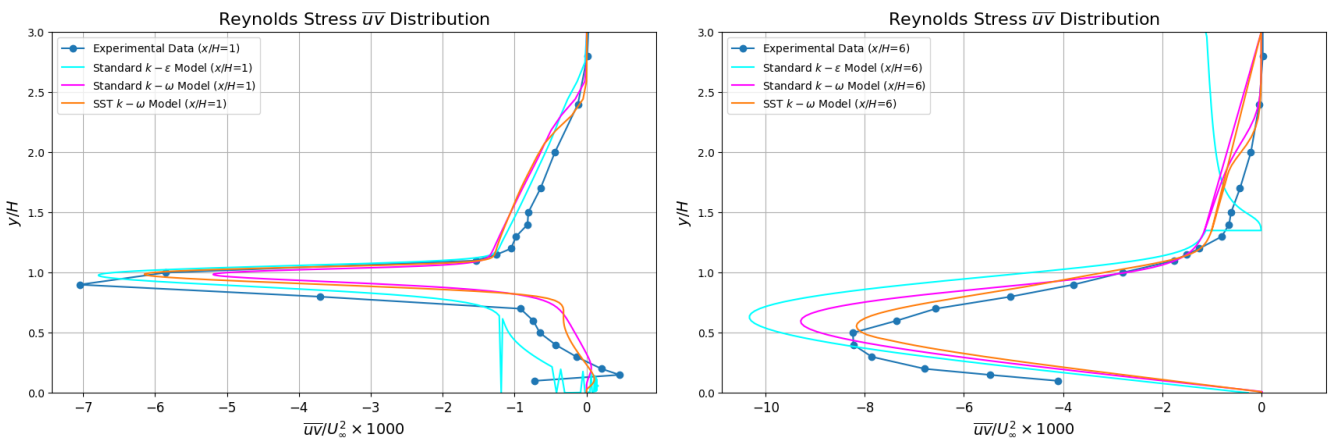


(a) Reynolds stress $\overline{vv}$ component distribution at $x/H = 1$ for (b) Reynolds stress $\overline{vv}$ component distribution at $x/H = 6$ for different turbulence models compared with experimental data. different turbulence models compared with experimental data.

A comparison of the Reynolds normal stress component $\overline{vv}$ at positions $x/H = 1$ and $x/H = 6$ reveals that the peak magnitude is significantly smaller at $x/H = 1$ than at $x/H = 6$. This behavior can be attributed to the unde-

veloped nature of wall-normal turbulence shortly after reattachment. Near the reattachment point, the turbulence is highly anisotropic, with streamwise fluctuations dominating. As the flow progresses downstream to $x/H = 6$, the turbulence becomes more isotropic and fully developed, resulting in stronger wall-normal velocity fluctuations and a corresponding increase in the $\overline{vv}$ stress component.

Results of $\overline{uv}$ component of Reynolds stress is presented in Fig. 6a and 6b. In both cases, the predicted distributions are consistently negative, which aligns with physical expectations, as the negative $\overline{uv}$ represents turbulent momentum transfer from high-speed to low-speed regions in the wall-normal direction. However, at $x/H = 6$, all turbulence models overpredict the magnitude of $\overline{uv}$, especially in the region away from the wall. This overestimation is likely due to the turbulence models generating excess turbulent viscosity, leading to an exaggerated shear stress. This issue is common in RANS-based models, which assume isotropic turbulence and often fail to accurately capture the complex stress recovery in reattached boundary layers.



(a) Reynolds stress $\overline{uv}$ component distribution at $x/H = 1$ for different turbulence models compared with experimental data. (b) Reynolds stress $\overline{uv}$ component distribution at $x/H = 6$ for different turbulence models compared with experimental data.

The Reynolds shear stress component $\overline{uv}$, representing turbulent momentum transport, shows distinct behavior at different streamwise locations. Experimental data indicates that the peak value of $\overline{uv}$ at $x/H = 6$ increases slightly, comparing to peak at $x/H = 1$, reflecting enhanced turbulent mixing as the flow redevelops after reattachment. Among the turbulence models evaluated, the SST $k-\omega$ model demonstrates the best agreement with experimental data at $x/H = 6$, accurately capturing both the peak magnitude and the shape of the shear stress distribution. This performance can be attributed to the SST model's ability to blend the near-wall accuracy of the $k-\omega$ formulation with the free-shear behavior of the $k-\varepsilon$ model, making it particularly well-suited for flows involving separation and reattachment.

Conclusion

This study has evaluated the performance of three widely used RANS-based turbulence models in simulating a backward-facing step flow, focusing on key flow characteristics such as pressure coefficient, skin friction, velocity profiles, and Reynolds stress distributions at strategic streamwise locations. The results reveal that each model exhibits strengths in different regions of the flow, highlighting the importance of selecting a turbulence model suited to the specific flow features of interest.

Overall, the SST $k-\omega$ model emerges as the most balanced and robust option across all flow regions, combining the strengths of both $k-\omega$ and $k-\varepsilon$ formulations. It is particularly recommended for engineering applications involving separation, reattachment, and strong pressure gradients, where accurate prediction of turbulent behavior is critical.

References

[1] David M. Driver and Harold L. Seegmiller. 2dbfs: 2d backward facing step. `https://turbmodels.larc.nasa.gov/backstep_val.html`, 1985.

[2] Inc. ANSYS. *Standard k-epsilon Model*, 2025.

[3] Inc. ANSYS. *Standard k-omega Model*, 2025.

[4] Inc. ANSYS. *SST k-omega Model*, 2025.